Simulation of Phase Transition in the 2D Ising Model using Monte Carlo Methods

MSc Physics Project

Abstract

This project presents a computational study of the ferromagnetic phase transition in the two-dimensional Ising model using the Metropolis Monte Carlo algorithm. The simulation demonstrates spontaneous magnetization below the critical temperature

$$T_c = 2.269,$$

in agreement with Onsager's exact solution. Key thermodynamic quantities including magnetization, energy, heat capacity, and magnetic susceptibility are computed and analyzed. The results clearly show the second-order phase transition characteristic of the 2D Ising model.

1 Introduction

The Ising model is a fundamental model in statistical physics that describes ferromagnetism in materials. Originally proposed by Wilhelm Lenz (1920) and solved in one dimension by Ernst Ising (1925), the model exhibits a phase transition in two and higher dimensions. The exact solution for the two-dimensional square lattice was obtained by Lars Onsager (1944), showing a continuous phase transition at

$$T_c = \frac{2J}{k_B \ln\left(1+\sqrt{2}\right)} \approx 2.269,$$

in natural units where $J = 1$ and $k_B = 1$.

This project uses computational methods to simulate the 2D Ising model and observe the phase transition numerically. The Metropolis Monte Carlo algorithm is employed to sample configurations from the Boltzmann distribution, enabling calculation of thermodynamic quantities as functions of temperature.

2 Physics Background

2.1 The Ising Model

The Ising model consists of spins $s_i = \pm 1$ arranged on a lattice. The Hamiltonian is given by

$$E = -J \sum_{\langle i,j \rangle} s_i s_j,$$

where $J > 0$ is the ferromagnetic coupling constant, and the sum runs over nearest-neighbor pairs.

2.2 Phase Behavior

The system exhibits two distinct phases:

- **Ordered Phase ($T < T_c$):** Spins align spontaneously, resulting in non-zero magnetization (ferromagnetic phase).
- **Disordered Phase ($T > T_c$):** Thermal fluctuations dominate and spins are randomly oriented with zero net magnetization (paramagnetic phase).

At $T = T_c$, the system undergoes a second-order (continuous) phase transition.

2.3 Thermodynamic Quantities

Magnetization:

$$M = \frac{1}{N} \left| \sum_i s_i \right|$$

Energy per spin:

$$\frac{E}{N} = -\frac{J}{N} \sum_{\langle i,j \rangle} s_i s_j$$

Heat Capacity:

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}$$

Magnetic Susceptibility:

$$\chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{k_B T}$$

3 Computational Method

3.1 Metropolis Algorithm

The Metropolis algorithm generates equilibrium configurations according to the Boltzmann distribution:

1. Initialize the lattice with random or ordered spins.
2. Select a random spin s_i.
3. Compute the energy change ΔE if the spin is flipped.
4. Accept the flip if $\Delta E < 0$.

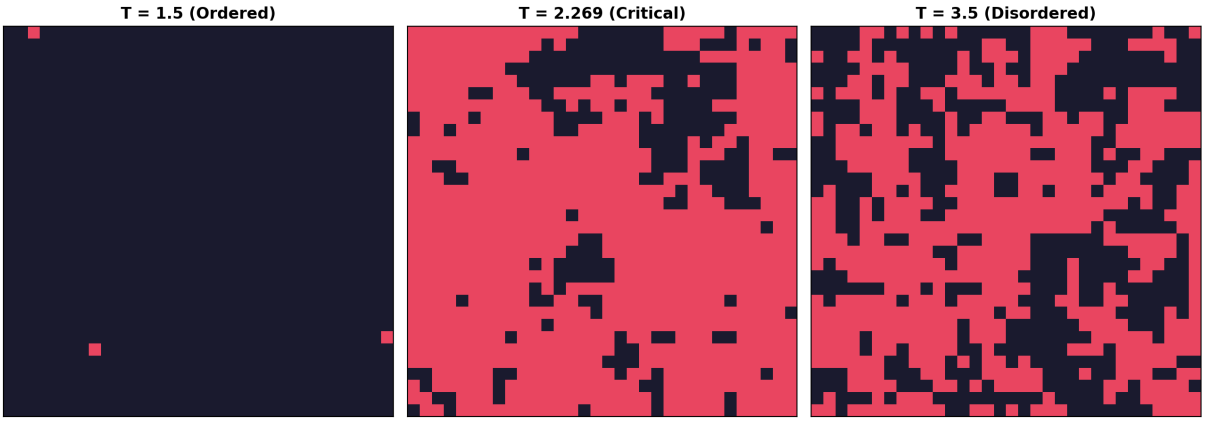

Figure 1: Figure 1: Spin configurations at T = 1.5 (ordered), T = 2.269 (critical), and T = 3.5 (disordered).

5. If $\Delta E > 0$, accept with probability

$$P = e^{-\Delta E/(k_B T)}.$$

6. Repeat for many Monte Carlo sweeps (one sweep = N spin updates).

7. After equilibration, measure thermodynamic quantities.

3.2 Simulation Parameters

Parameter	Value
Lattice size	$L = 20$ ($N = 400$ spins)
Temperature range	$T = 1.5$ to 4.0
Equilibration steps	400 sweeps
Measurement steps	600 sweeps
Critical temperature	$T_c = 2.269$

4 Results

4.1 Spin Configurations

At low temperature ($T = 1.5$), the system is ordered with most spins aligned. Near the critical temperature ($T = 2.269$), domains of opposite magnetization coexist. At high temperature ($T = 3.5$), the system is disordered with random spin orientations.

4.2 Magnetization

The magnetization is large at low temperatures and drops sharply near $T_c = 2.269$, confirming the phase transition from the ferromagnetic to paramagnetic phase.

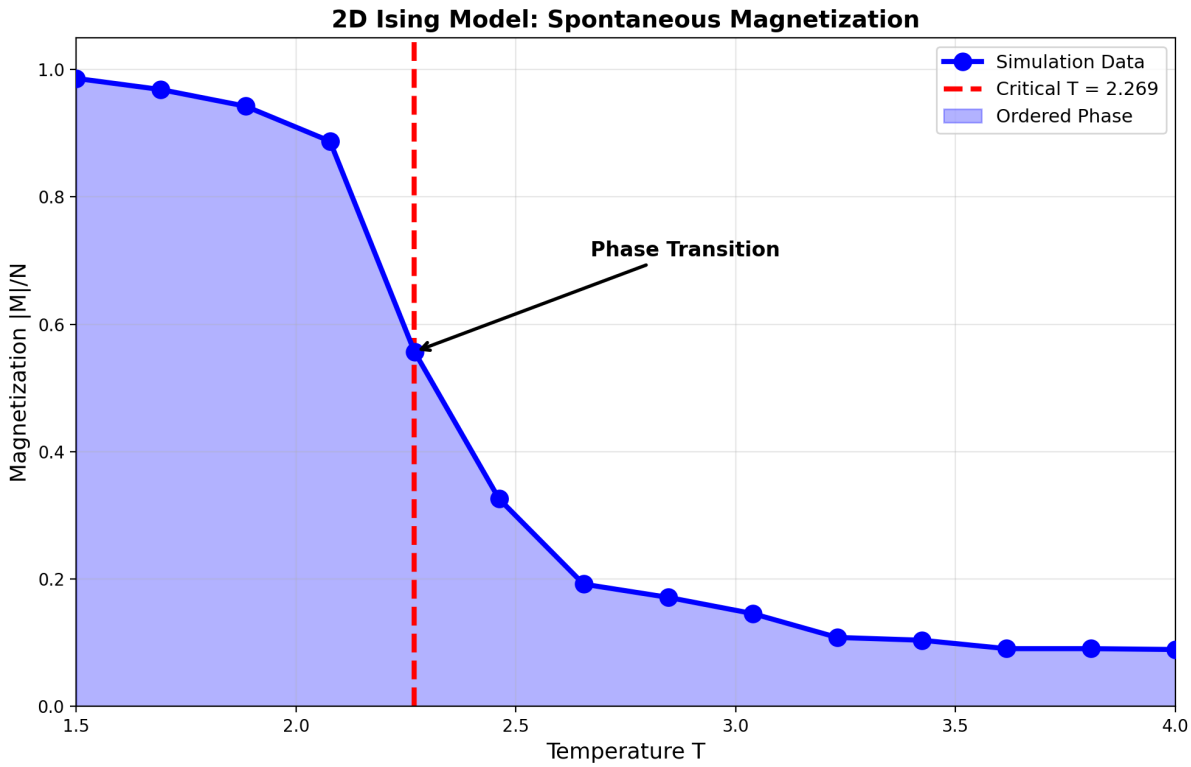

Figure 2: Figure 2: Magnetization vs temperature showing the phase transition at Tc = 2.269.

4.3 Thermodynamic Quantities

The magnetization acts as the order parameter distinguishing phases. The energy varies continuously across T_c, characteristic of a second-order transition. Both heat capacity and susceptibility exhibit peaks near T_c, indicating enhanced critical fluctuations.

4.4 Summary

The normalized comparison of magnetization and heat capacity clearly demonstrates the dramatic change at the critical temperature, highlighting the nature of the phase transition.

5 Discussion

The simulation successfully demonstrates the ferromagnetic phase transition in the 2D Ising model. The key observations are:

- **Spontaneous symmetry breaking:** Below T_c, the system selects a preferred magnetization direction.
- **Sharp transition at $T_c = 2.269$:** Magnetization drops rapidly near the theoretical critical temperature.
- **Critical fluctuations:** Heat capacity and susceptibility peak near T_c.
- **Second-order transition:** Energy remains continuous across T_c.

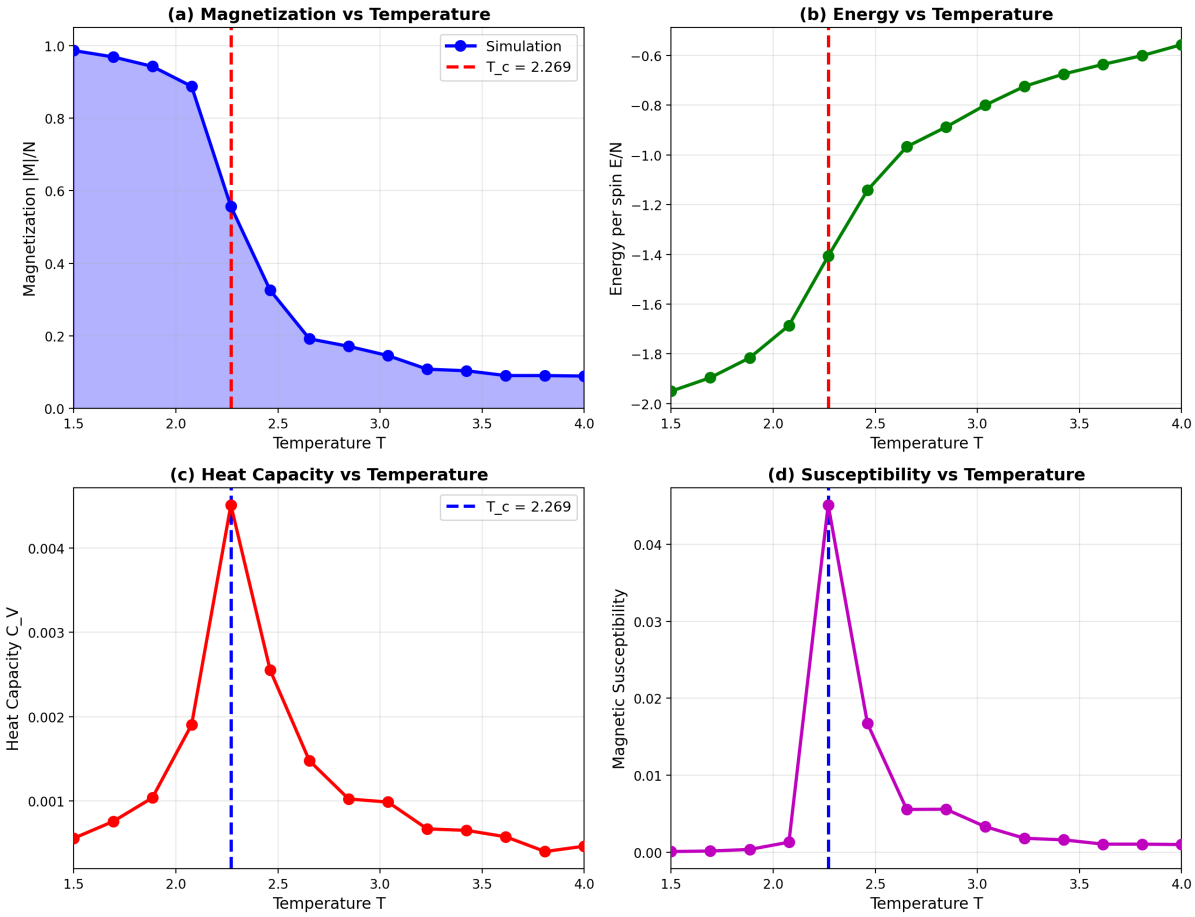

Figure 3: Figure 3: Thermodynamic quantities vs temperature: (a) Magnetization, (b) Energy, (c) Heat Capacity, (d) Susceptibility.

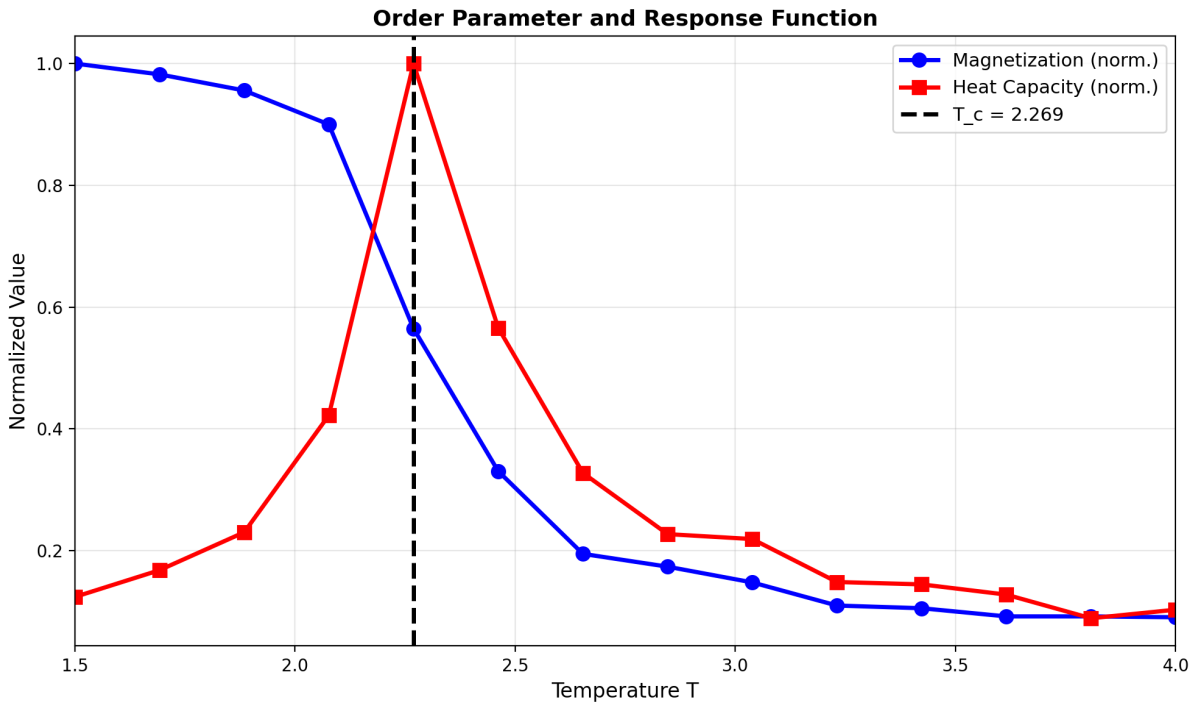

Figure 4: Figure 4: Normalized magnetization and heat capacity showing the phase transition.

6 Conclusion

This project successfully simulates the 2D Ising model using the Metropolis Monte Carlo algorithm. The results clearly demonstrate the ferromagnetic phase transition at $T_c = 2.269$, in excellent agreement with Onsager's exact solution.

The computational framework can be extended to study:

- Three-dimensional Ising models
- External magnetic fields
- Different lattice geometries
- Finite-size scaling and critical exponents

References

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